

Question description

A company has n employees with certain salaries. Your task is to keep track of the salaries and process queries.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq p_i \leq 10^9$
- $1 \leq k \leq n$
- $1 \leq x \leq 10^9$
- $1 \leq a \leq b \leq 10^9$

Input

The first input line contains two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$.

The next line has n integers $p_1, p_2, \dots, p_n$: each employee's salary.

After this, there are q lines describing the queries. Each line has one of the following forms:

- $! k x$: change the salary of employee k to x
- $? a b$: count the number of employees whose salary is between $a \dots b$

Output

Print the answer to each $?$ query.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5 3
3 7 2 2 5
? 2 3
! 3 6
? 2 3

EXPECTED OUTPUT

3
2

Test Case 2

INPUT (STDIN)

5 3
31 17 12 12 15
? 12 31
! 13 6
? 2 13

EXPECTED OUTPUT

5
3

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int compare(const void
*a,const void *b)

Test Case 2

KEYWORD

void update(int i,int
n,int x)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

1

Test Case 2

TOKEN COUNT

575

Test Case 3

NLOC

65